

First overview of governing equations

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1 A note on notations

1.1 Material derivative

The following equations are all equivalent, which can be shown algebraically.

$$\begin{aligned}\frac{d}{dt} &= \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z} \\ &= \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \\ &= \frac{\partial}{\partial t} + \mathbf{u} \cdot \text{div}()\end{aligned}$$

1.2 Tensors

$$\tau_{ij} \frac{\partial u_i}{\partial x_j} = \boldsymbol{\tau} : \nabla \mathbf{u}$$

where

$$\nabla \mathbf{u} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}$$

The $:$ is called the double contraction operator or the matrix inner product. It contracts two tensors into a scalar. Since τ is the symmetric stress tensor,

2 Starting equations (partial differential equations)

2.1 Energy equation

2.1.1 CRB

CRB: Cushman-Roisin and Beckers, 2011

$$\rho C_v \frac{dT}{dt} + p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = k_T \nabla^2 T \quad (1)$$

I refer to Schubert et al., 2001 for the conversion into the constant-pressure heat capacity variant of this equation.

2.1.2 Jarvis & McKenzie, 1980

jamc80 Jarvis and McKenzie, 1980:

$$\rho C_p \left(\frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p} \frac{\partial P}{\partial t} + \mathbf{u} \cdot [\nabla T - (\nabla T)_s] \right) = \nabla \cdot (K \nabla T) + H + \tau_{ij} \frac{\partial u_i}{\partial x_j} \quad (2)$$

2.1.3 Gassmüller et al., 2020

Gassmüller: Gassmüller et al., 2020

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) - \nabla \cdot (k \nabla T) = \rho H + \tau : \dot{\varepsilon} + \alpha T (\mathbf{u} \cdot \nabla p) + Q_L \quad (3)$$

2.1.4 Schubert et al., 2001

Schubert et al., 2001 presents three thermmal energy equations:

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \rho H \quad (4)$$

For an incompressible Newtonian fluid (eq. 6.9.13 + 6.9.14):

$$\rho C_v \frac{DT}{Dt} = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (5)$$

with $\frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2$ the viscous dissipation function.

In terms of C_p instead of C_v (for a compressible fluid) (eq. 6.9.16):

$$\rho C_p \left(\frac{DT}{Dt} - \left(\frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} \right) = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (6)$$

with (eq. 6.9.15, 6.9.17):

$$\begin{aligned} \left(\frac{\partial T}{\partial p} \right)_s &= \frac{\alpha T}{\rho C_p} \\ \left(\frac{DT}{Dt} \right)_s &= \left(\frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} = \frac{\alpha T}{\rho C_p} \frac{Dp}{Dt} \end{aligned}$$

2.2 Momentum equation

2.2.1 CRB

CRB: Cushman-Roisin and Beckers, 2011 per component

$$x : \rho \left(\frac{du}{dt} + f_* w - f v \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau^{xx}}{\partial x} + \frac{\partial \tau^{xy}}{\partial y} + \frac{\partial \tau^{xz}}{\partial z} \quad (7a)$$

$$y : \rho \left(\frac{dv}{dt} + f u \right) = -\frac{\partial p}{\partial y} + \frac{\tau^{xy}}{\partial x} + \frac{\partial \tau^{yy}}{\partial y} + \frac{\partial \tau^{yz}}{\partial z} \quad (7b)$$

$$z : \rho \left(\frac{dw}{dt} - f_* u \right) = -\frac{\partial p}{\partial z} - \rho g + \frac{\tau^{xz}}{\partial x} + \frac{\tau^{yz}}{\partial y} + \frac{\partial \tau^{zz}}{\partial z} \quad (7c)$$

2.2.2 Jarvis & McKenzie, 1980

jamc80, eq. 4 Jarvis and Mckenzie, 1980:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \rho \mathbf{g} + \frac{\partial}{\partial x_j} \tau_{ij} \quad (8)$$

2.2.3 Gassmüller et al., 2020

Gassmüller Gassmüller et al., 2020:

$$-\nabla \cdot \tau + \nabla p = \rho \mathbf{g} \quad (9)$$

2.2.4 Schubert et al. 2001

$$\rho \frac{du_i}{dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \quad (10)$$

In 2D, the momentum equation components are:

$$\rho \left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \quad (11)$$

$$\rho \left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \quad (12)$$

Note that the assumption is made that gravity acts in the positive y-direction. Section 6.14?

2.3 Mass equation (continuity equation) (trivial)

jamc80, eq. 3 Jarvis and Mckenzie, 1980:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (13)$$

CRB Cushman-Roisin and Beckers, 2011:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \quad (14)$$

Gassmüller Gassmüller et al., 2020:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (15)$$

schubert et al. (2001) Schubert et al., 2001

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_1}(\rho u_i) = \frac{1}{\rho} \frac{D\rho}{Dt} + \frac{\partial u_1}{\partial x_1} = 0 \quad (16)$$

where $\frac{D}{Dt}$ represents the material derivative (??). For incompressible flow, $\frac{D\rho}{Dt} = 0$, so eq ?? reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \quad (17)$$

2.4 Starting equations (constitutive equations)

2.5 Initial and boundary conditions

3 Appendix 1: tables of constants and reference/scaling values used

4 Schubert et al., 2001

This document/section has a different approach to structure the derivation of the ALA: this is a summary of the derivation of the ALA done by Schubert et al., 2001.

4.1 Mass equation/continuity equation/conservation of mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_1}(\rho u_i) = \frac{1}{\rho} \frac{D\rho}{Dt} + \frac{\partial u_1}{\partial x_1} = 0 \quad (18)$$

where $\frac{D}{Dt}$ represents the material derivative (??). For incompressible flow, $\frac{D\rho}{Dt} = 0$, so eq ?? reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \quad (19)$$

For the special case of an incompressible fluid, $\frac{D\rho}{Dt} = 0$, so ?? reducesto

Table 1: Constants used by Schubert et al., 2001

Constant	Value	Name
k_b	0	Bulk viscosity
α		thermal expansivity
χ_T		isothermal compressibility
c_v		specific heat at constant volume
c_p		specific heat at constant pressure
χ_a		adiabatic compressibility
k		thermal conductivity
H		rate of internal heat production per unit mass
K_a		adiabatic bulk modulus
K_T		isothermal bulk modulus

Table 2: Reference values used by Schubert et al., 2001

Symbol	Value	Description
ΔT		Characteristic temperature difference driving thermal convection
χ_{T_r}		Representative isothermal compressibility
α_r		Representative thermal expansivity
u_r	$\frac{k_r}{\rho_r c_{p_r} b}$	Velocity scale
μ_r	10^{21} Pa s	Representative viscosity
χ_a		Adiabatic compressibility
b		Depth of convecting region
p_r	$\frac{\mu_r u_r}{b}$	Scale for dynamic pressure variations
k_r	$4 \text{ W m}^{-1} \text{ K}^{-1}$	Reference value thermal conductivity
c_{p_r}	$1 \text{ kJ kg}^{-1} \text{ K}^{-1}$	Reference specific heat at constant pressure

4.2 Thermal energy equations

Schubert et al., 2001 presents three thermal energy equations:

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \rho H \quad (20)$$

For an incompressible Newtonian fluid (eq. 6.9.13 + 6.9.14):

$$\rho C_v \frac{DT}{Dt} = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (21)$$

with $\frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2$ the viscous dissipation function.

In terms of C_p instead of C_v (for a compressible fluid) (eq. 6.9.16):

$$\rho C_p \left(\frac{DT}{Dt} - \left(\frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} \right) = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (22)$$

Table 3: Dimensionless parameters and their values for the Earth’s mantle

Parameter	Definition	Value	Description
Pr	$\frac{\mu_r c_{p_r}}{k_r}$	2.5×10^{23}	Prandtl number
M ²	$\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}$	10^{-33}	Mach number (squared)
ϵ	$\alpha_r \Delta T_r$		Fractional density change due to temperature variations

with (eq. 6.9.15, 6.9.17):

$$\begin{aligned}
\left(\frac{\partial T}{\partial p}\right)_s &= \frac{\alpha T}{\rho C_p} \\
\left(\frac{DT}{Dt}\right)_s &= \left(\frac{\partial T}{\partial p}\right)_s \frac{Dp}{Dt} = \frac{\alpha T}{\rho C_p} \frac{Dp}{Dt} \\
\rho \frac{du_i}{dt} &= -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i
\end{aligned} \tag{23}$$

4.3 Momentum equations

In 2D, the momentum equation components are:

$$\rho \left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \tag{24}$$

$$\rho \left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \tag{25}$$

Note that the assumption is made that gravity acts in the positive y-direction. Section 6.14?

References

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- Gassmöller, R., Dannberg, J., Bangerth, W., Heister, T., & Myhill, R. (2020). On formulations of compressible mantle convection. *Geophysical Journal International*, 221(2), 1264–1280. <https://doi.org/10.1093/gji/ggaa078>
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